JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SANKAGIRI LORRY OWENER'S ASSOCIAATI

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

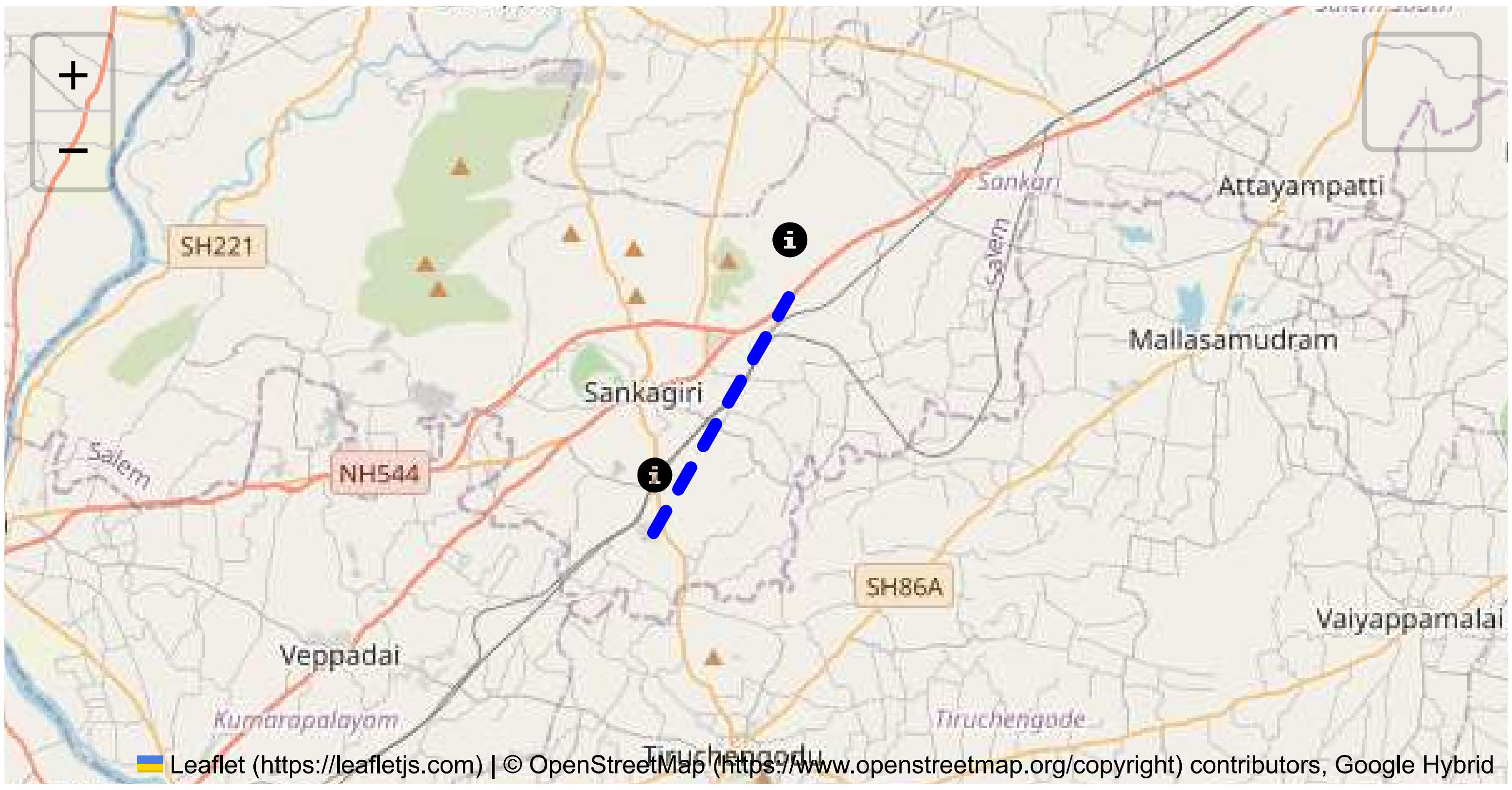
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

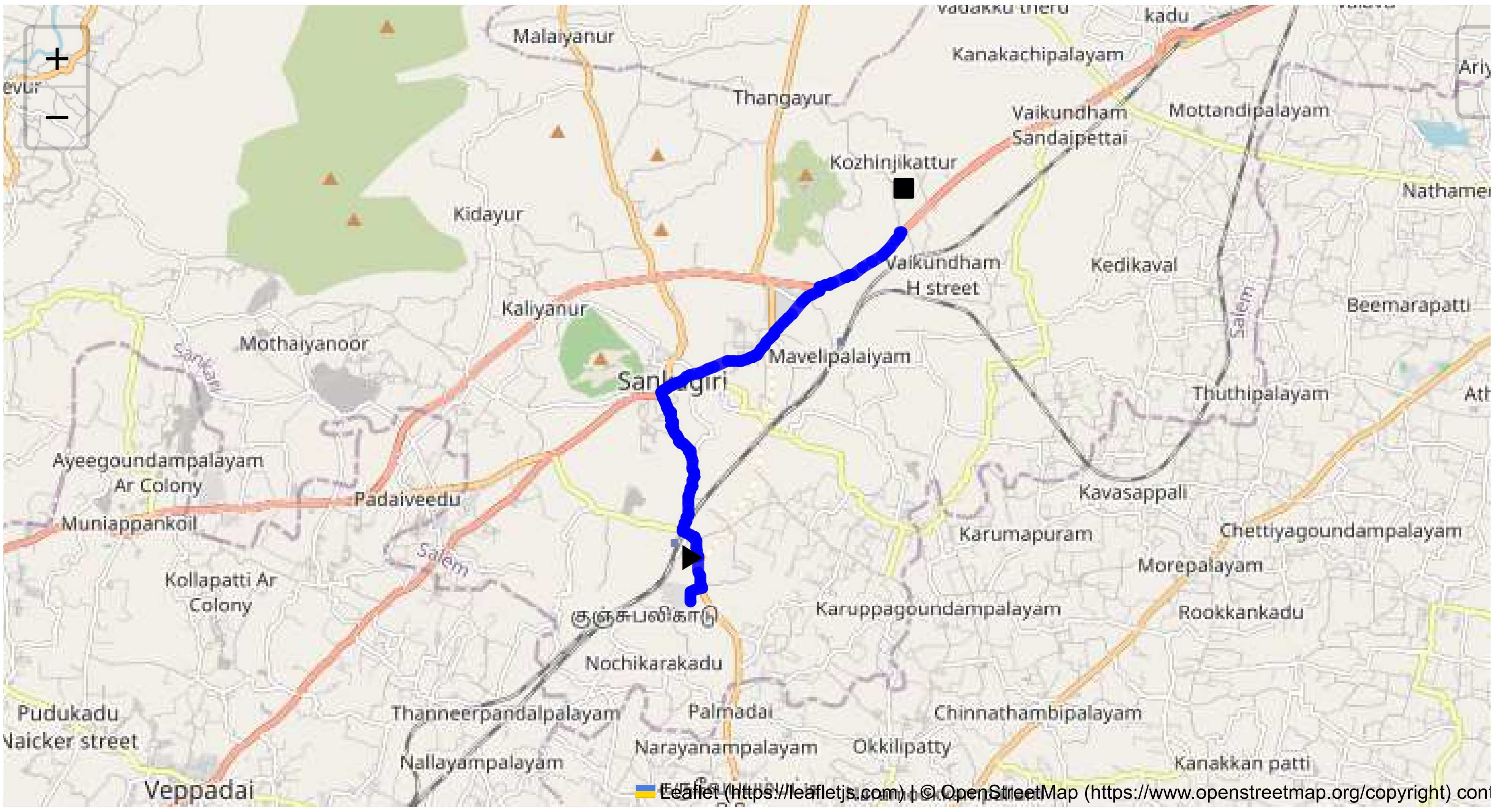
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 10.30 km
Estimated Duration: 0.3 hours
Adjusted Duration (Heavy Vehicle): 0.4 hours
Start: (11.4381, 77.8734)
End: (11.501725, 77.910476)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to GW26+H48, Service Road, Avarangampalayam covers approximately 10.3 kilometers, typically including local roads, possibly intersecting with regional highways. The route mainly runs through rural and semi-urban areas in Tamil Nadu.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions**: Tamil Nadu generally experiences tropical weather, with hot summers, moderate monsoons, and mild winters.
- **Hazards**: During monsoon season (June to September), heavy rains can cause waterlogging or flooding, possibly affecting road conditions and visibility. The summer months can bring heatwaves and occasional thunderstorms.

3. Traffic Patterns, Peak Hours, and Congestion-Prone Areas

- **Traffic Patterns**: Typically, rural routes experience moderate traffic, but nearby urban areas may encounter congestion.
- **Peak Hours**: Morning (8 AM - 10 AM) and evening (5 PM - 7 PM) are peak times due to local commuters.
- **Congestion-Prone Areas**: Any intersections with major highways or roads leading into larger townships or markets can be bottlenecks.

4. Assessment of Road Quality and Infrastructure

- **Road Quality**: Generally, roads in rural Tamil Nadu can range from well-paved to poorly maintained. Potholes and narrow lanes are common issues.
- **Infrastructure**: Limited street lighting, inadequate signage, and lack of dedicated lanes for heavy vehicles or hazardous materials transport are notable weaknesses.

5. Suggestions for Alternative Routes for Emergencies

- In emergencies, consider detouring via nearby NH and SH roads which are likely to be better maintained and have higher availability of services and facilities.
- Consulting real-time navigation apps for live updates on road conditions during an emergency can be beneficial.

6. Local Regulations Affecting Hazardous Material Transport

- **Regulations**: Adherence to guidelines on the movement of hazardous materials, such as having appropriate signage on the vehicle, drivers trained and licensed for hazardous material, and specific time windows for transportation on certain routes.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Incidents**: Limited specific data may be available, but generally, incidents in rural areas involve vehicle accidents due to poor road conditions or driver error.

8. Environmental Considerations and Sensitive Areas

- **Environmental Areas**: Routes through agricultural zones may have more stringent regulations to prevent contamination from hazardous spills.
- **Sensitive Areas**: Presence of schools, markets, or wildlife areas should be noted as requiring extra caution.

9. Communication Coverage, Noting Potential Dead Zones

- **Coverage**: Rural areas might experience patchy mobile network coverage, especially with lesser-known service providers. Ensuring devices are fully charged and secondary means of communication, such as satellite phones, could be considered.

10. Estimated Emergency Response Times for Different Route Segments

- **Response Times**: In urban-semi urban segments, response times could be within 30 minutes, while in rural sections, it might extend to over an hour due to distance from major medical or fire response facilities.

12. An Overall Summary of Risk Assessment

The route generally poses moderate risk primarily due to variable road conditions, potential weather hazards during the monsoon, and communication coverage limitations. Proactive measures such as route planning during favorable weather, preparation for communication dead zones, and the driver's thorough knowledge of regulations around hazardous material transport are essential. Continuous monitoring of weather forecasts and real-time traffic updates can mitigate risks, ensuring safer transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	High	11.48369, 77.88781	15 KM/Hr
7	Turn	Medium	11.48370, 77.88787	30 KM/Hr
8	Turn	Medium	11.49149, 77.89601	30 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Blind Spot	Blind Spot	11.50108, 77.91039	10 KM/Hr
11	Blind Spot	Blind Spot	11.50119, 77.91025	10 KM/Hr
12	Turn	High	11.50150, 77.91052	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.48369, 77.88781

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48370, 77.88787

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49149, 77.89601

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.50108, 77.91039

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.50119, 77.91025

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.50150, 77.91052